

CPNP Laboratory Project



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Project Topic: Theme 2

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Theme 2:

It is considered a commercial building with 3 levels. Use network address 172.27.0.0/16 for intranet network, network address 210.2.2.64/27 for DMZ and network address 210.2.2.32/27 for access to the exterior. Design four VLANs (one for each floor and one for traffic management). VTP protocol will be used to configure VLANs. Redundancy will be provided through network cabling and configuration. Host addresses will be assigned dynamically using a single DHCP server located in the VLAN corresponding to the first floor. The minimum number of users served by each VLAN is 200. HTTP, FTP, DNS and Mail servers will be placed in the DMZ and will have public addresses. Web domain name will include the student's name. To ensure connectivity static routes will be configured. Access to the exterior will be done using NAT on the router that controls the DMZ, on the following range of public addresses: 210.2.2.35-210.2.2.62.

Connecting to the ISP will be achieved through an Ethernet interface having the address 210.2.2.34/27. ISP address is 210.2.2.33/27. Internet network will be simulated through a server and a computer.

For securing network devices the following configurations will be performed: users on different privilege levels will be defined, passwords will be encrypted, remote configuration will be done only through ssh, VTP protocol will be secured.

Did not implement this requirement:

Two additional security measures will be presented and implemented.

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Session I

We are starting to build our configuration. We have a commercial building with 3 floors, so we will have a MainRouter (Router0), connected to MainSwitch (Switch0). Then, a Switch for each of the three floors.

We need to subnet 172.27.0.0/16.

We will create Vlans for each floor: Vlan10 for FirstFloor, Vlan20 for SecondFloor, Vlan30 for ThirdFloor. We start by configuring Vlan20, since the SecondFloor switch also has Server0 attached.

172.27.20.0/24

172.27.20.1 - gateway

172.27.20.2 - DHCP

172.27.20.10 → hosts

```
!  
interface FastEthernet0/10  
    switchport access vlan 20  
    switchport mode access  
!  
interface FastEthernet0/11  
    switchport access vlan 20  
    switchport mode access  
,
```

We then configure Vlan10 and Vlan30 as well.

```
.  
interface FastEthernet0/10  
    switchport access vlan 30  
    switchport mode access  
,  
.  
interface FastEthernet0/10  
    switchport access vlan 10  
    switchport mode access  
,
```

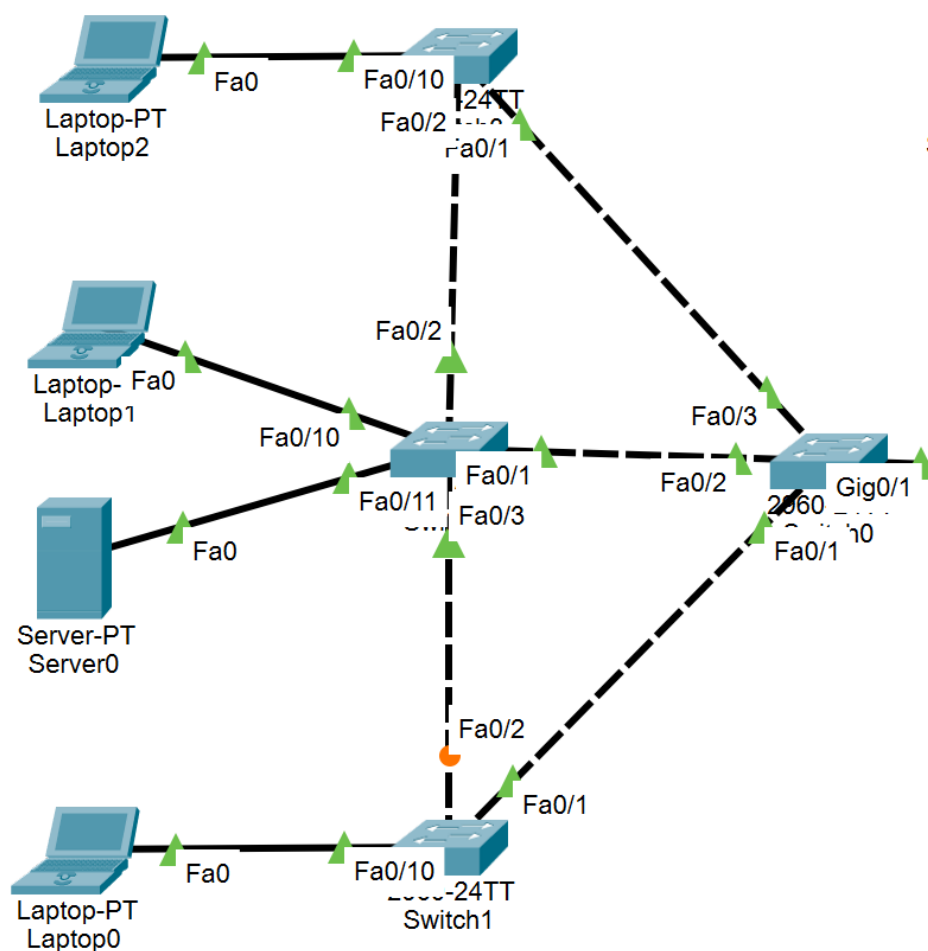
Vlan99 will be used for traffic management.

IP Configuration

Interface	FastEthernet0
IP Configuration	
☒ DHCP ☐ Static	
IPv4 Address	172.27.30.10
Subnet Mask	255.255.255.0
Default Gateway	172.27.30.1
DNS Server	210.2.2.68

We also configure the host addresses, using a DHCP server located in the VLAN of the FirstFloor.

The configuration can be tested using the **show running config** command, and the connections between the elements will be tested using **ping** in the Command Prompt.



Session II

In this session, we configure the HTTP, FTP, DNS and Mail servers that will be placed in the DMZ (demilitarized zone). The DMZ is a network used to connect hosts that provide an interface to an untrusted external network - the internet, usually - while keeping the internal network separated and isolated.

```
interface GigabitEthernet0/1
  switchport access vlan 2
  switchport mode access
!
interface GigabitEthernet0/2
  switchport access vlan 2
  switchport mode access
!
interface Vlan1
  no ip address
  shutdown
!
interface Vlan2
  ip address 210.2.2.70 255.255.255.224
!
ip default-gateway 210.2.2.65
!
```

We setup the DMZ switch

To setup the four servers (HTTP, FTP, DNS and Mail), we need to assign them IP addresses:

- HTTP - 210.2.2.66, subnet 255.255.255.224
- FTP - 210.2.2.67
- DNS - 210.2.2.68
- Mail - 210.2.2.69

With default gateway 210.2.2.65, and DNS server 210.2.2.68.

We can test the connections by using the **ping** command in the Command Prompt of other devices, to this device.

This is the HTTP page:

Web Browser
X

<
>
URL
http://www.denisa.ro|
Go
Stop

Denisa

Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

Quick Links:

[A small page](#)
[Copyrights](#)
[Image page](#)
[Image](#)

Also, we use the DNS server to translate the other servers to a name:

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type A Record

Address

Add
Save
Remove

No.	Name	Type	Detail
0	ftp.denisa.ro	A Record	210.2.2.67
1	mail.denisa.com	A Record	100.0.0.2
2	mail.denisa.ro	A Record	210.2.2.69
3	www.denisa.ro	A Record	210.2.2.66

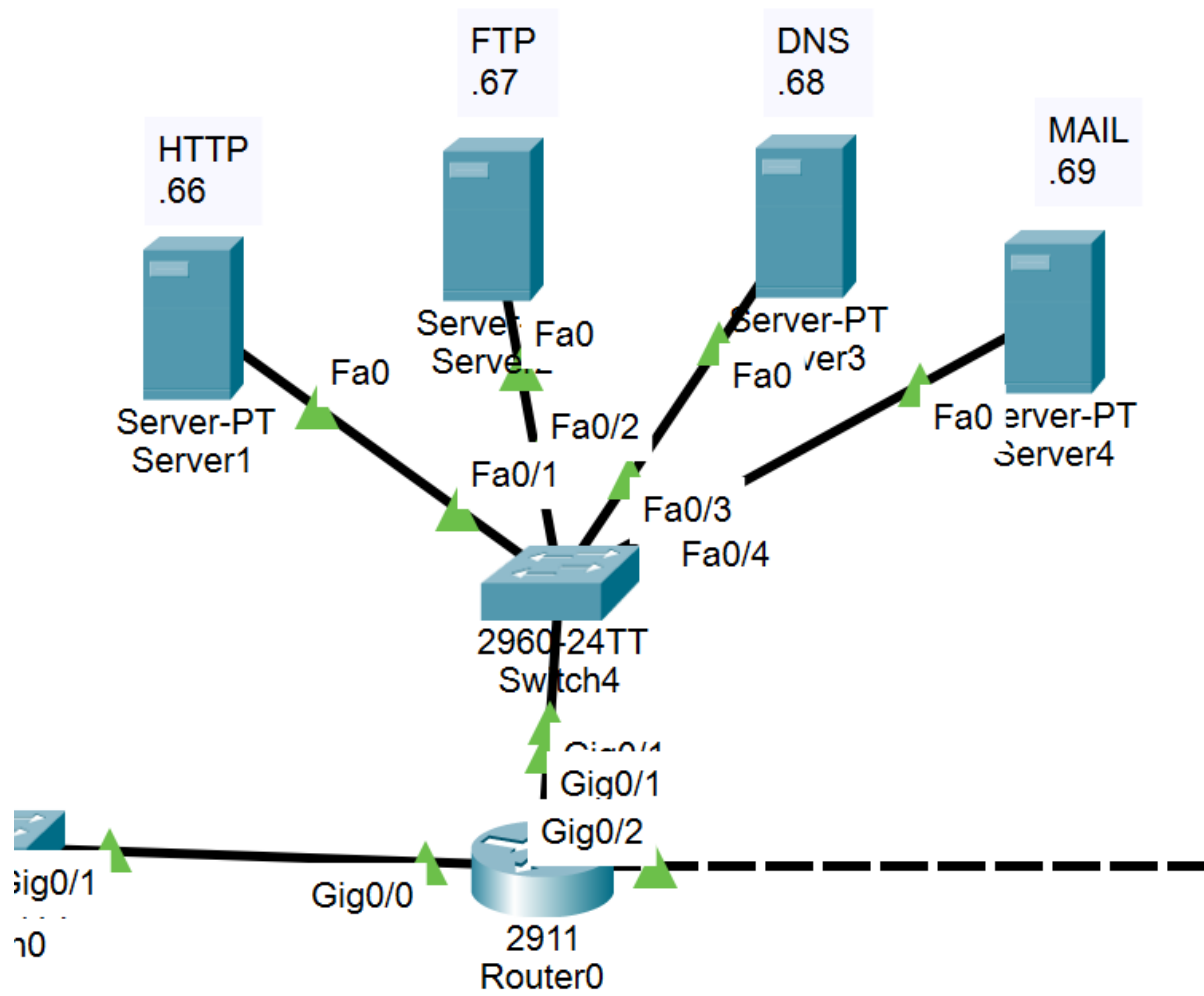
And as for the Mail server, we create to users: denisa1 and denisa2. We will test the server by passing emails between the users on Laptop2:

MAIL BROWSER
X

Mails

Compose
Reply
Receive
Delete
Configure Mail

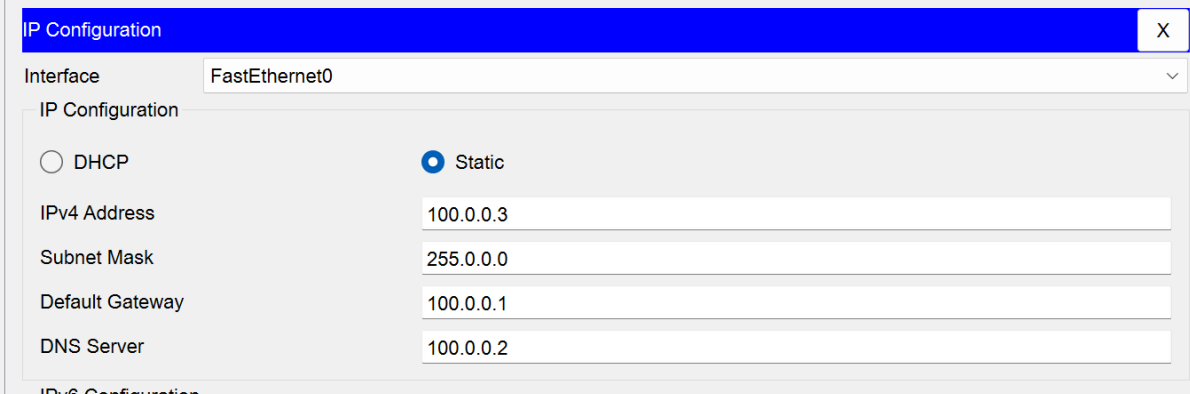
	From	Subject	Received
1	denisa3@mail.denisa.com	RE: testt	Thu Mar 16 2023 15:50:35
2	denisa1@mail.denisa.ro	test32	Thu Mar 16 2023 15:46:37
3	denisa2@mail.denisa.ro	test	Thu Mar 16 2023 15:26:24
4	denisa1@mail.denisa.ro	aaa	Thu Mar 16 2023 15:24:41



Session III

In the third session, we manage the ISP (Internet Service Provider), that is Router1. The Laptop3 is our client.

We need to configure the router, switch, server, and laptop by assigning them their IP addresses. The configuration for the laptop, for example:



IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 100.0.0.3

Subnet Mask: 255.0.0.0

Default Gateway: 100.0.0.1

DNS Server: 100.0.0.2

IPv6 Configuration

The server is also the Mail, DHCP and DNS server, so we need to properly configure it:

DNS

DNS Service ☒ On ☐ Off

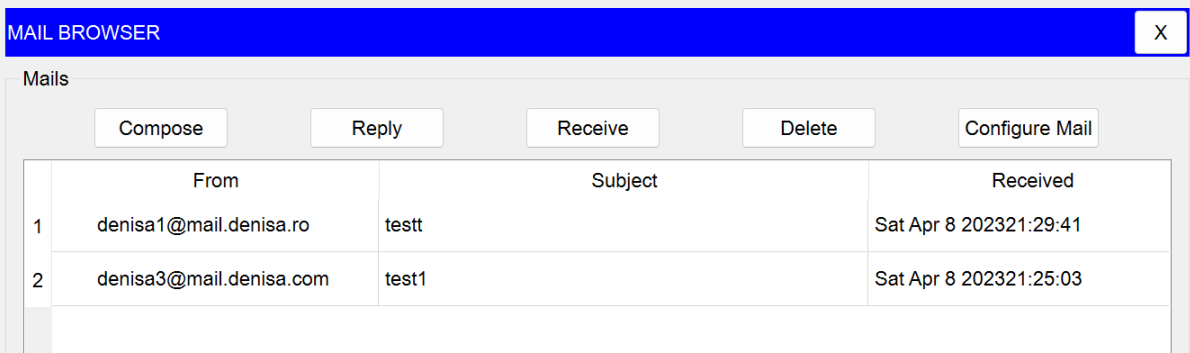
Resource Records

Name: Type: A Record

Address:

No.	Name	Type	Detail
0	mail.denisa.com	A Record	100.0.0.2
1	mail.denisa.ro	A Record	210.2.2.69

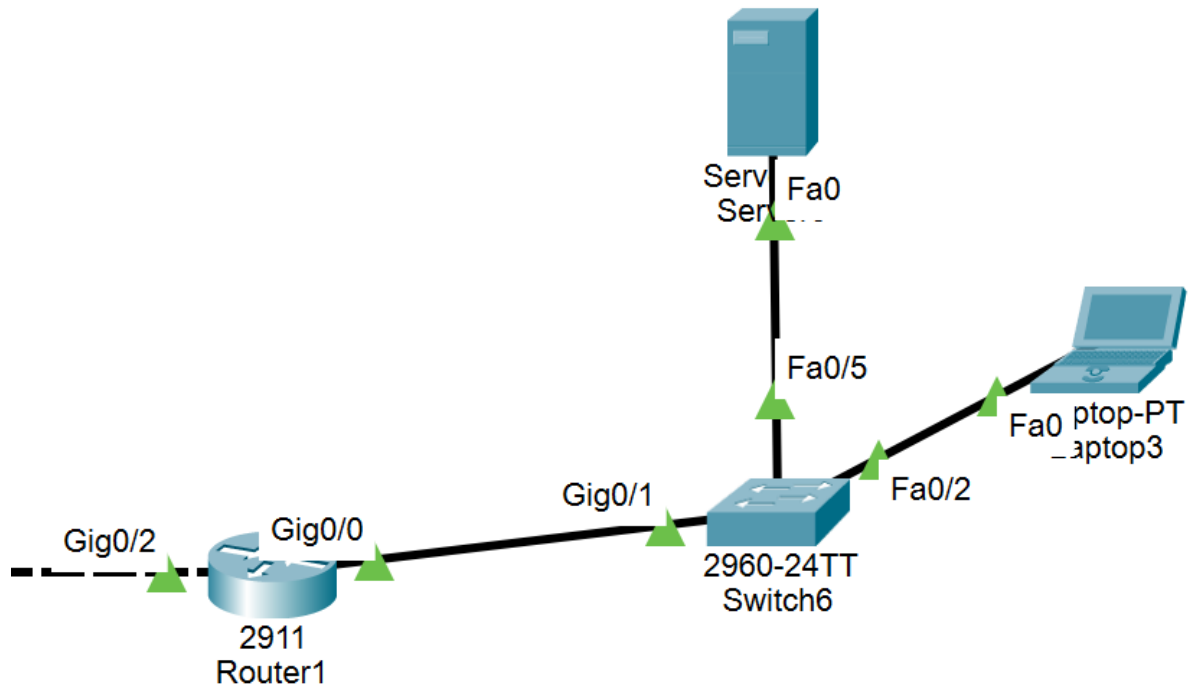
We also add the denisa3 user for the mail, and test it by passing some emails on Laptop3.



MAIL BROWSER [X]

Mails

	From	Subject	Received
1	denisa1@mail.denisa.ro	testt	Sat Apr 8 2023 21:29:41
2	denisa3@mail.denisa.com	test1	Sat Apr 8 2023 21:25:03



Session IV

When displaying **MainRouter>**, the privilege level is just 1.

With the **enable** command, we enter the privileged mode, and we have more permissions for the commands. The existing commands also have more parameters. The privilege level is now 15.

We are now creating 3 users:

1. **username denisa1 privilege 1 password denisa1**
2. **username denisa7 privilege 7 password denisa7**
3. **username denisa15 privilege 15 secret denisa15**

To test what we have done:

do sh run

```
username denisa1 password 0 denisa1
username denisa15 privilege 15 secret 5 $1$mERr$uwqGUQskXXxOd8tJuGdzX/
username denisa7 privilege 7 password 0 denisa7
```

A digital certificate is possible due to the asymmetrical cryptography methods. To generate the two necessary keys, since they are related to the domain server, we look in the DNS server we have created in the previous session, where we have the domain server **denisa.ro**. To get the name of the keys to be **MainRouter.denisa.ro**, we use the command:

ip domain-name denisa.ro

We generate the private and public keys with:

crypto key generate rsa, with 2048 bits. The public key will be given to everybody, and the private key will be kept secret.

line vty 0 4 → it will accept 5 simultaneous connections, the 6th one won't be accepted

transport input ssh → accept SSH connections

login local → the login will be done with the local database

We now test the equipment we have just configured:

do sh run

do show users → only 1 user is connected

On a laptop, we will verify the IP address and mask. Test the connection to the router: **ping 210.2.2.65**. We connect to SSH to the MainRouter: **ssh -l denisa1 210.2.2.65**, then enter the password when prompted. We have the privilege 1 and can enter commands for the MainRouter.

```
MainRouter>show users
```

Line	User	Host(s)	Idle	Location
0 con 0		idle	00:02:36	
*390 vty 0	denisa1	idle	00:00:00	

Interface	User	Mode	Idle	Peer Address
MainRouter>				

To have the privilege 15, we can connect with SSH to the **denisa15** user, or enter the enable command. But we cannot enter the enable command remotely, since we have a password set.

enable secret denisa15

We get back to the command prompt of the laptop, enter **enable** and the **denisa15** password, and now we have privilege 15.

copy running-config startup-config, to save the configuration.

We move now to Switch4 from DMZ.

1. **username denisa1 privilege 1 password denisa1**
2. **username denisa7 privilege 7 password denisa7**
3. **username denisa15 privilege 15 password denisa15**
4. **enable password denisa15**
5. **ip domain-name denisa.ro**
6. **crypto key generate rsa** for DMZ.denisa.ro
7. **line vty 0 4**
8. **transport input ssh**
9. **login local**

To test: **do show run**

```
interface GigabitEthernet0/1
  switchport access vlan 2
  switchport mode access
!
interface GigabitEthernet0/2
  switchport access vlan 2
  switchport mode access
!
interface Vlan1
  no ip address
  shutdown
!
interface Vlan2
  ip address 210.2.2.70 255.255.255.224
!
ip default-gateway 210.2.2.65
!
!
```

We go to Laptop1, Command Prompt, and enter:

ssh -l denisa1 210.2.2.70

enable → enter the password denisa15

Now we have remote access to **DMZ#**

```
DMZ#sh run
Building configuration...

Current configuration : 2694 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname DMZ
!
enable password denisa15
!
!
!
ip domain-name denisa.ro
!
username denisa1 privilege 1 password 0 denisa1
username denisa15 privilege 15 password 0 denisa15
username denisa7 privilege 7 password 0 denisa7
!
!
--More--
```

To encrypt all plain text passwords:

service password-encryption

```
DMZ(config)#do show run
Building configuration...

Current configuration : 2730 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname DMZ
!
enable password 7 08254940000A044647
!
!
!
ip domain-name denisa.ro
!
username denisa1 privilege 1 password 7 08254940000A0446
username denisa15 privilege 1 password 7 08254940000A044647
username denisa7 privilege 1 password 7 08254940000A0440
!
!
--More--
```

```
username denisa1 privilege 1 password 7 08254940000A0446
username denisa15 privilege 1 password 7 08254940000A044647
username denisa2 privilege 2 password 7 08254940000A0445
username denisa7 privilege 1 password 7 08254940000A0440
```